

Cluster Analysis with SAP HANA

Cluster analysis with SQLNotebook and Calculation Views

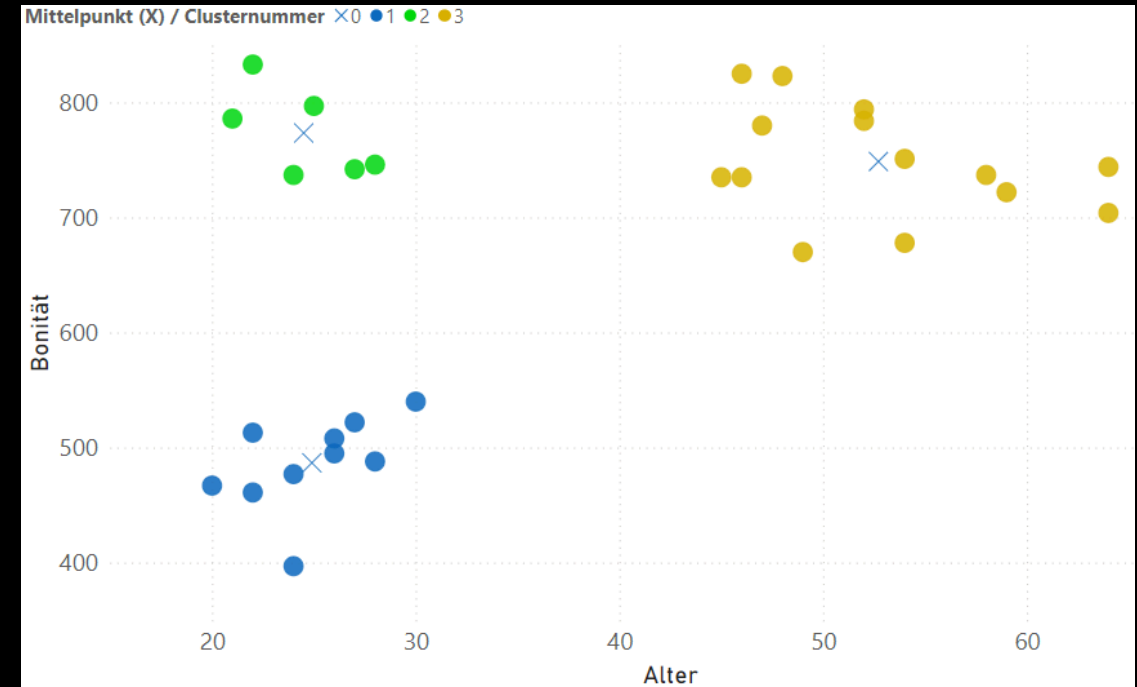
About me (Benedict Baur)

- Freelance SAP consultant and developer
- Focus area:
 - SAP Business Warehouse (BW/4)
 - SAP HANA Modeling with Calculation Views
 - Machine Learning (SAP HANA, Python)
- Trainer for Calculation Views (for Brandeis Consulting)
- Contact me:
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Use Case: Customer Data Segmentation

- Given: Data Set of customers of a company (*Bank, Insurance, Telecommunication* etc.)
- Data Set contains several attributes (*age, location, salary, sales amount*) and information about behavior of customers
- Goal: Segment data set in groups (*clusters*) of similar customers
 - Similarity is calculated using attributes of the input data set



Cluster Analysis in SQLScript

- Use SQLScript for Cluster Analysis with PAL Procedures
- Execute in SQLNotebook in Business Application Studio
- Store results in table for further analysis

The screenshot shows the SQLNotebook interface in Business Application Studio. The top bar indicates the current context: 'SQLNotebookClustering > db > Cluster_Analysis.notebook > ...'. Below this, there are tabs for '+ Code' and '+ Markdown', and buttons for 'Run All', 'Clear All Outputs', and a menu icon. The main editor area contains a SQLScript snippet: `FROM CHURN WHERE BALANCE > 0;` followed by a call to the PAL_KMEANS procedure: `CALL _SYS_AFL.PAL_KMEANS(:lt_churn, :lt_parameter, lt_result, lt_centers, lt_model, lt_statistics, lt_placeholder);`. Below the code editor, the execution results are displayed. A message states: 'Statement: 'DO BEGIN DECLARE lt_parameter TABLE(NAME VARCHAR (50), INT_VALUE INTEGER, ...' result :'. Below this message is a table with four columns: CUSTOMERID, CLUSTER_ID, DISTANCE, and SLIGHT_SILHOUETTE. The table contains six rows of data. At the bottom right of the table, there is a link to 'Show 15 More'.

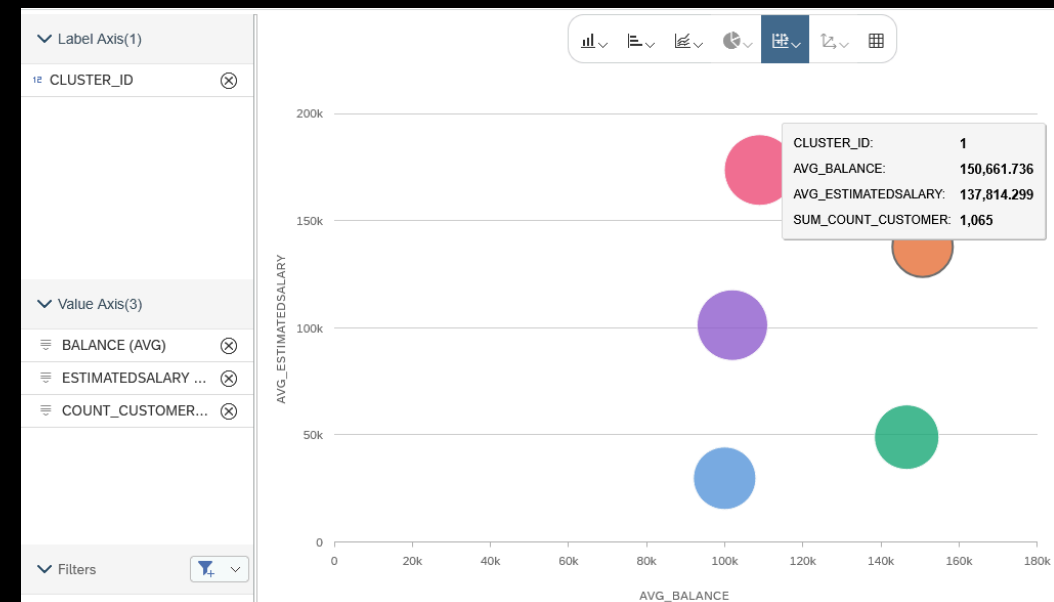
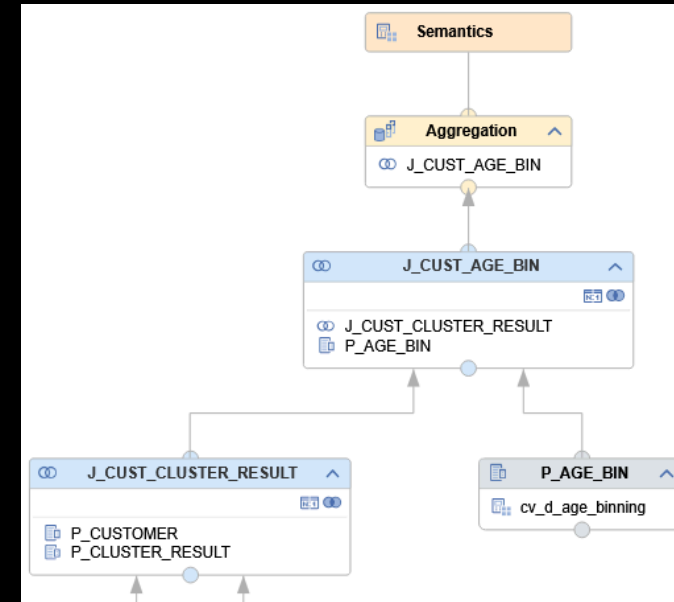
CUSTOMERID	CLUSTER_ID	DISTANCE	SLIGHT_SILHOUETTE
15647311	4	21435.576171875	0.6752055864444403
15619304	1	25477.380859375	0.5685373761180738
15737888	4	32283.3203125	0.12356077897554602
15574012	3	24281.38671875	0.37481148147824067
15656148	4	22345.3515625	0.44329461880628146

▼ Show 15 More

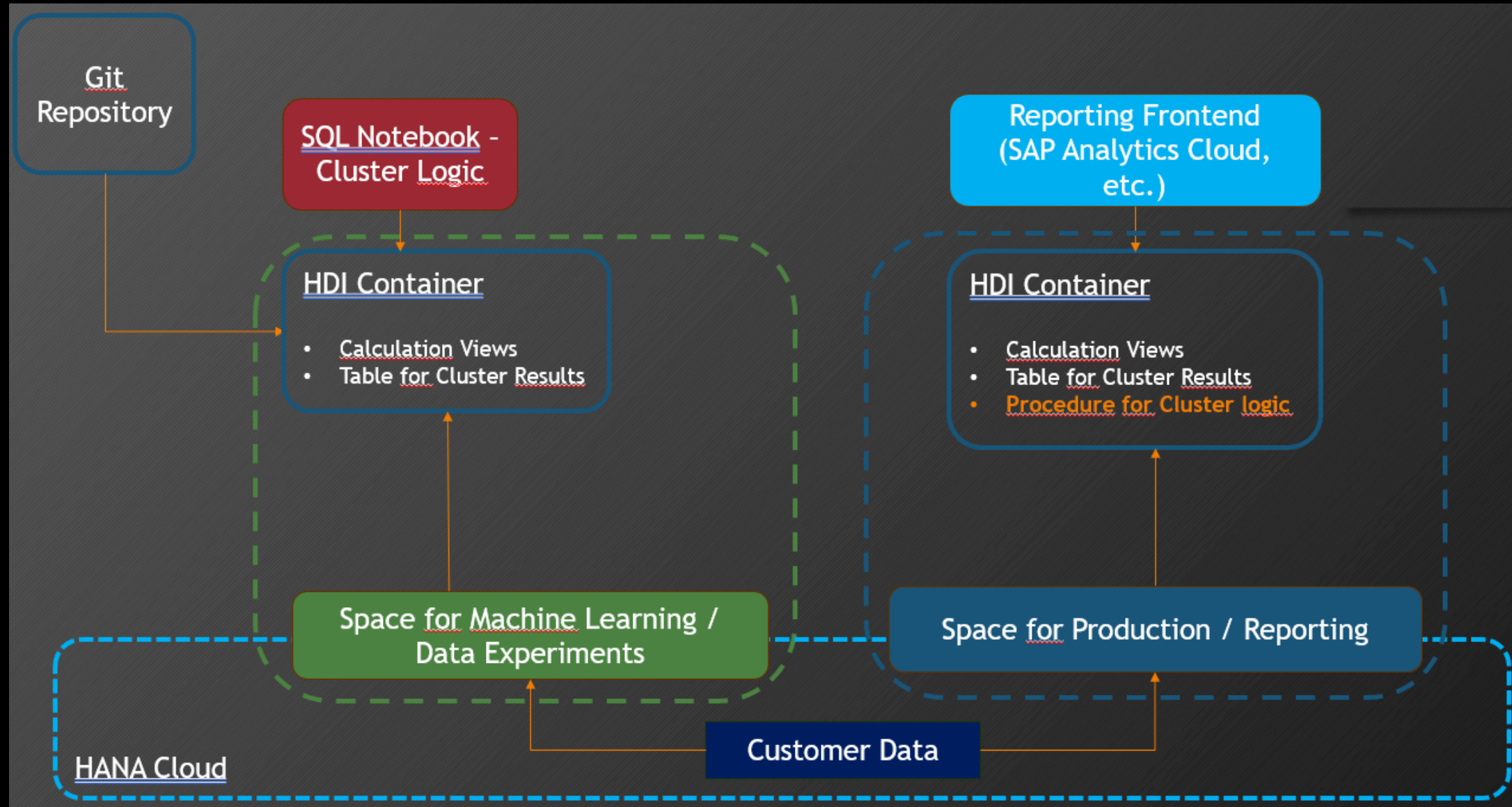
Evaluation of Cluster Result with Calculation Views

- Join customer attributes with cluster assignment of customers
- Use Calculation View modeling features for further evaluation of result
 - Window functions,
 - Aggregation,
 - Ranking, etc.
- Visualize in Data Preview

Advantage: Calculation View can be reused for several executions of Cluster Analysis.



HDI Container and Machine Learning



HDI container and Machine Learning

- Local development
- Every data scientist uses own container
- Store intermediate results
- Central deployment: Publish cluster analysis to reporting application
- Consumed by business users
- Access to external data using external services (central HANA tables, remote source, Data Sphere)